CAPSTONE PROJECT

Bathymetry AI Model

PRESENTED BY

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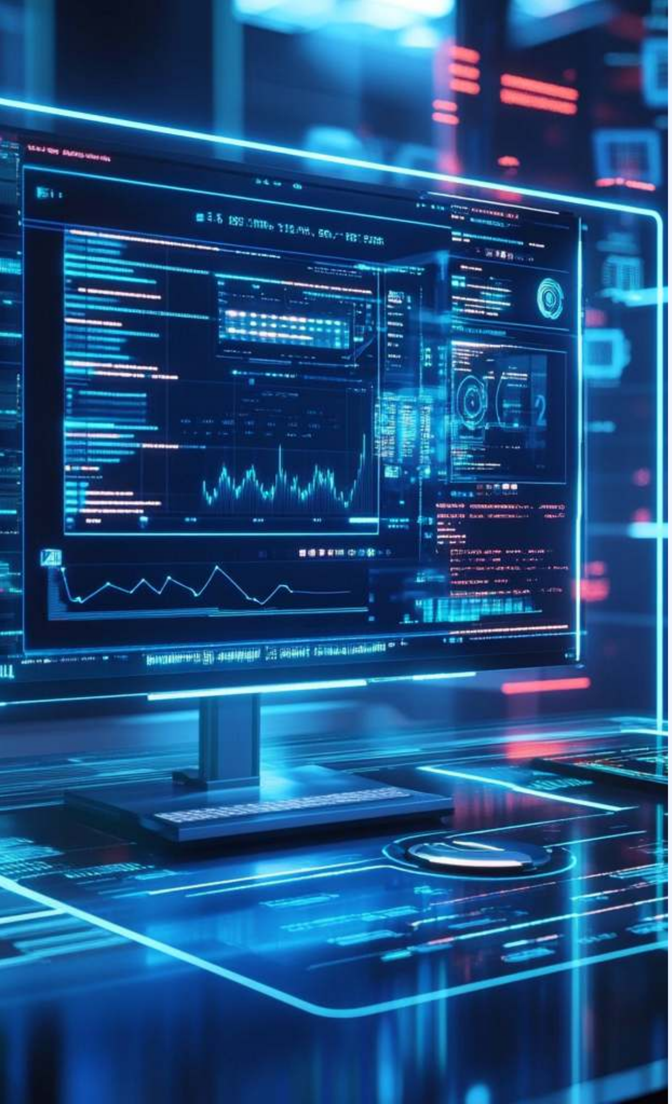
PINN-SSL-MMF: A Unified Deep Learning Approach for Bathymetry Mapping on MagicBathyNet



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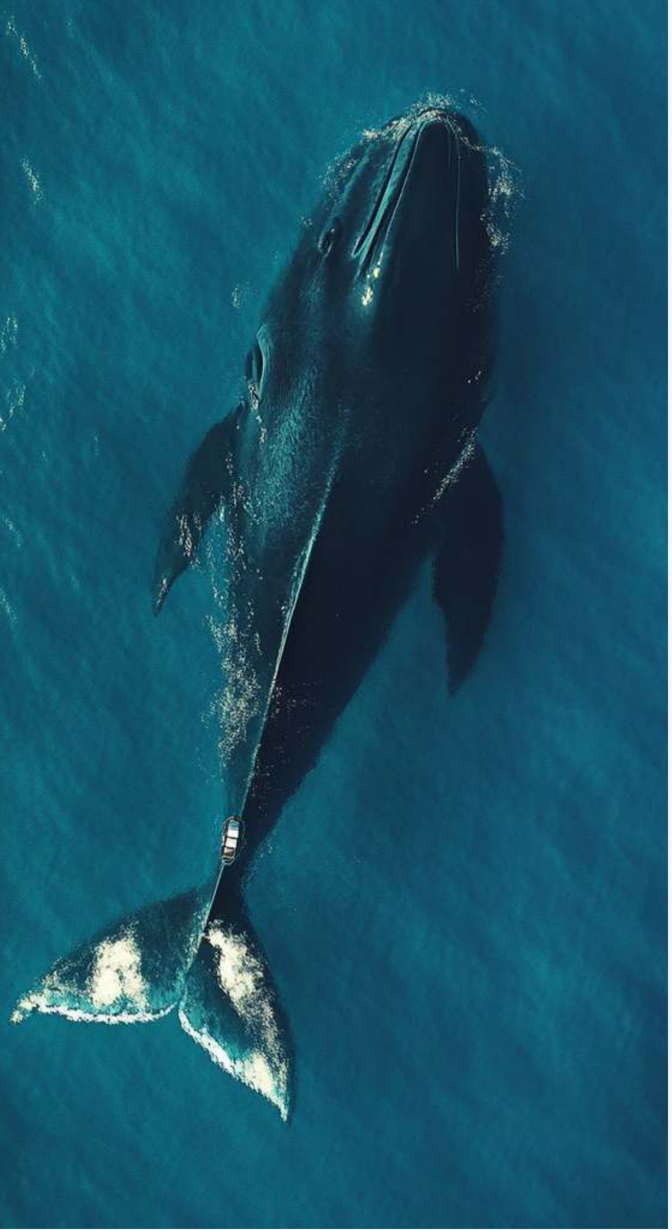
Problem Statement



Problem Statement

Accurate bathymetry mapping is essential for marine navigation, coastal management, and underwater exploration.

Traditional methods, which rely on costly sonar surveys, are often impractical for large-scale applications.

The integration of Satellite-Derived Bathymetry (SDB) presents a cost-effective alternative.

Challenges faced include limited labeled data, atmospheric noise, and inconsistencies due to environmental variations.

Addressing these issues is critical for improving navigation safety, environmental monitoring, military applications, and underwater exploration efficiency.

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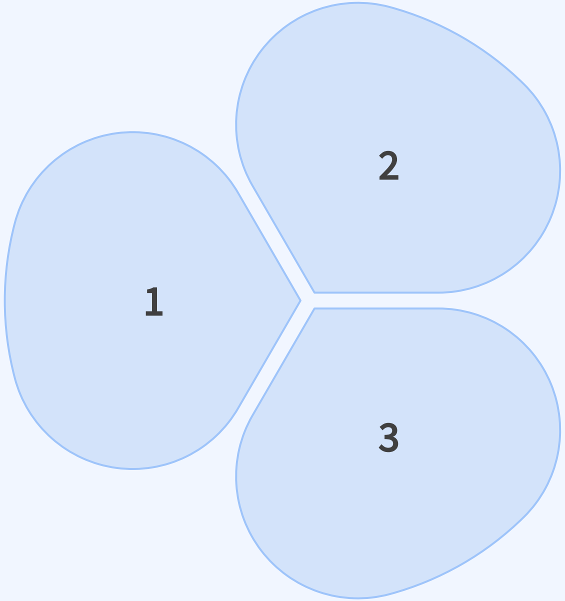
Proposed System / Solution

Proposed System / Solution

The proposed system PINN-SSL-MMF combines Physics-Inspired Neural Networks (PINN) for depth estimation, Self-Supervised Learning (SSL) for pre-training, and Multimodal Fusion (MMF) via various data sources, including aerial imagery, SPOT6 satellite data, and Sentinel-2 imagery.



The unique feature of this system lies in its ability to consolidate diverse data while adhering to physics-based constraints, achieving robust and accurate depth estimates across varying conditions.

The integration of self-supervised learning allows the model to leverage unlabeled data, enhancing its training efficacy without the need for extensive labeled datasets.

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System Development Approach (Technology Used)

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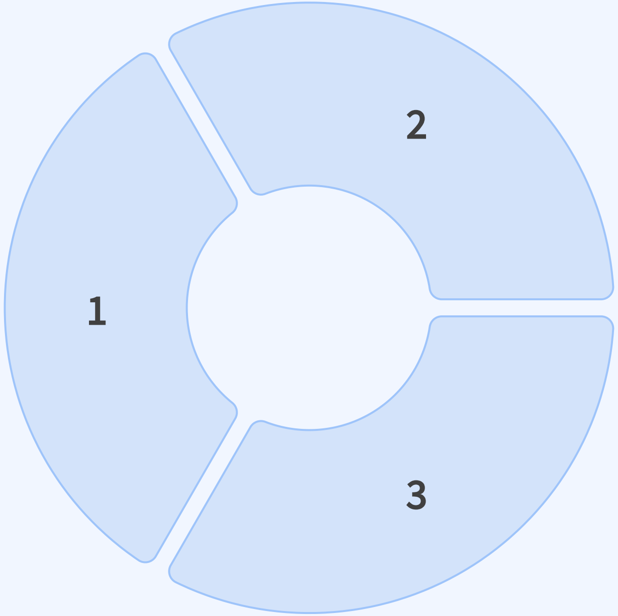
The development utilized Python as the primary programming language, with frameworks such as PyTorch for implementing deep learning models.



Techniques such as data normalization, augmentation, and mixed-precision training were employed to facilitate effective model performance.

Key technical decisions included the choice of the AdamW optimizer for training and the implementation of an attention-based fusion mechanism to synergize features from different sources effectively.

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Algorithm and Deployment

Algorithm and Deployment

The algorithm combines deep learning methodologies, focusing on a PINN architecture that incorporates physical laws, self-supervised learning techniques to enhance representation learning, and multimodal data processing.

The model was trained using a composite loss function that maintains data fidelity while conforming to physical realism.

The tested model was deployed on a cloud-based platform for efficient scalability, enabling real-time applications and large-scale bathymetry mapping with integer resolution support.

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Result (Output Image)

Result (Output Image) :

Modality	Agia Napa			Puck Lagoon		
	MSE	PSNR	SSIM	MSE	PSNR	SSIM
Sentinel-2	0.0046	35.3693	0.9144	0.0023	34.4504	0.7950
SPOT-6	0.0063	34.1982	0.9083	0.1268	24.8200	0.7348
Aerial	8.4924	-0.6848	0.1590	0.0728	30.6732	0.8240

Modality	Agia Napa (RMSE)	Agia Napa (Standard Deviation)	Agia Napa (MAE)	Puck Lagoon (RMSE)	Puck Lagoon (Standard Deviation)	Puck Lagoon (MAE)
Multimodal Combined	0.92 meters	0.91 meters	0.72 meters	2.65 meters	2.65 meters	1.44 meters

Result :

1 The results achieved include significant accuracy improvements over traditional and existing deep learning-based approaches, as evidenced by low RMSE scores and high SSIM values.

2 The model's output images demonstrated precise depth predictions, effectively illustrating its capability to reconstruct bathymetric features accurately.

3 Performance metrics during testing asserted ratios such as Mean Absolute Error (MAE) and Peak Signal-to-Noise Ratio (PSNR), highlighting the model’ s robustness across varying environmental conditions.

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Conclusion

Conclusion

1. The project presents a significant advancement in bathymetry mapping by integrating multiple advanced deep learning techniques into a unified framework.
2. It effectively addresses the limitations associated with traditional methods, offering a robust solution that enhances depth estimation accuracy, catering to diverse operational needs.
3. The contributions of this work may provide a crucial tool for marine navigation, environmental assessment, and military strategy development.

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Future Scope

Future Scope

Future enhancements may include the extension of the framework to accommodate hyperspectral data for enhanced depth estimation.

Scalability options could include additional integrations with emerging satellite data sources and real-time processing capabilities.

Long-term vision entails the deployment of the system in various practical applications such as underwater exploration and marine conservation efforts.

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References

References

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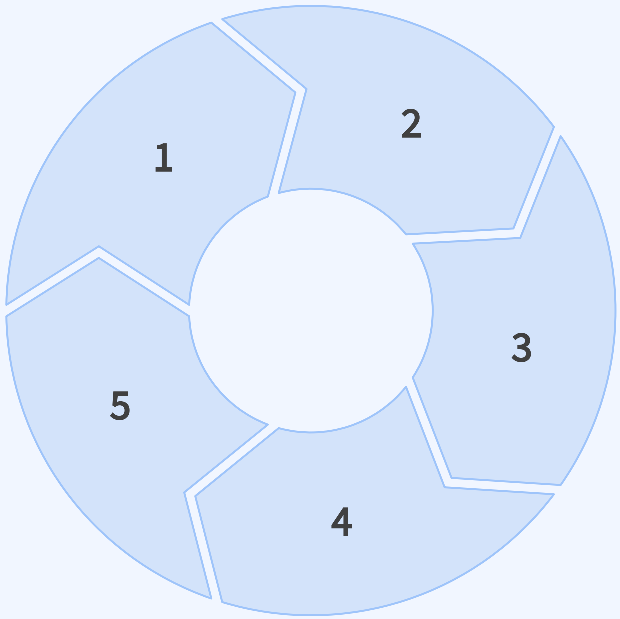

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GITHUB LINK : LINK

https://github.com/mynameisluckyji/Bathymetry-AI-Model

Thank You